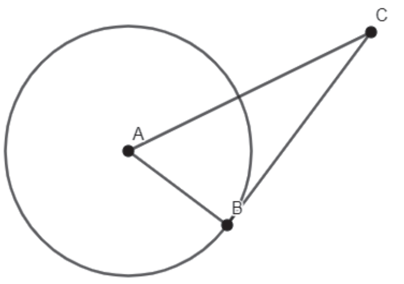
Geometry
Unit 12
Lesson 5 Practice

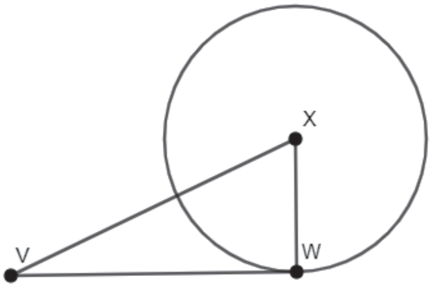
Name____________________
Date____________________
Period____________________

1. Circle $A$ is shown with point $C$ outside the circle, where point $B$ is on the circle, $AC = 26$, $CB = 24$, and $BA = 10$. Verify if $\overline{BC}$ is tangent to circle $A$ using Pythagorean Theorem.




Circle $X$ on the right has $\overline{VW}$ as a tangent segment where $W$ is the point of tangency. $VX = 17$ and $VW = 15$.

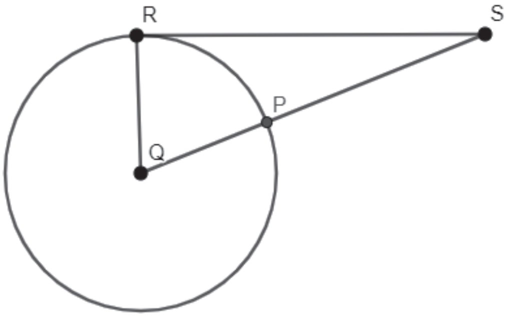
2. What is the length of $\overline{XW}$?




3. What is the area of $\Delta XWV$?

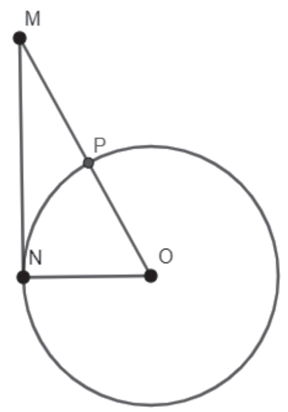
In Circle $Q$, $\overline{RS}$ is tangent to the circle at point $R$, $QR = 6$, and $PS = 4$.

4. What is the length of $\overline{RS}$?




5. What is the area of $\Delta QRS$?

In Circle $O$, $\overline{MN}$ is a tangent segment whose point of tangency is point $N$. $MN = 8$ and $MP = 2$.

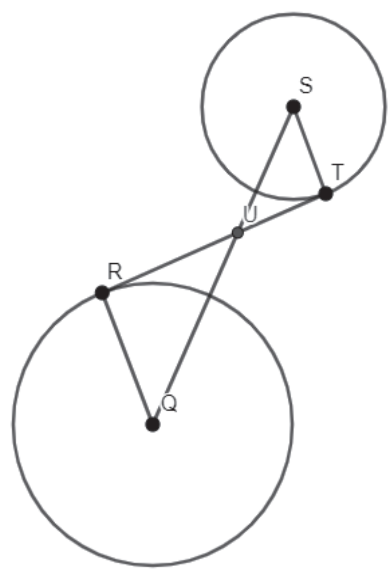



6. What is the length of $\overline{ON}$?

7. What is the length of $\overline{MO}$?

8. What is the perimeter of $\Delta MNO$?

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle $Q$ and Circle $S$. $QR = 5$, $RU = 12$, and $TU = 3$.




9. What is the length of $\overline{ST}$?

10. What is the length of $\overline{QS}$?